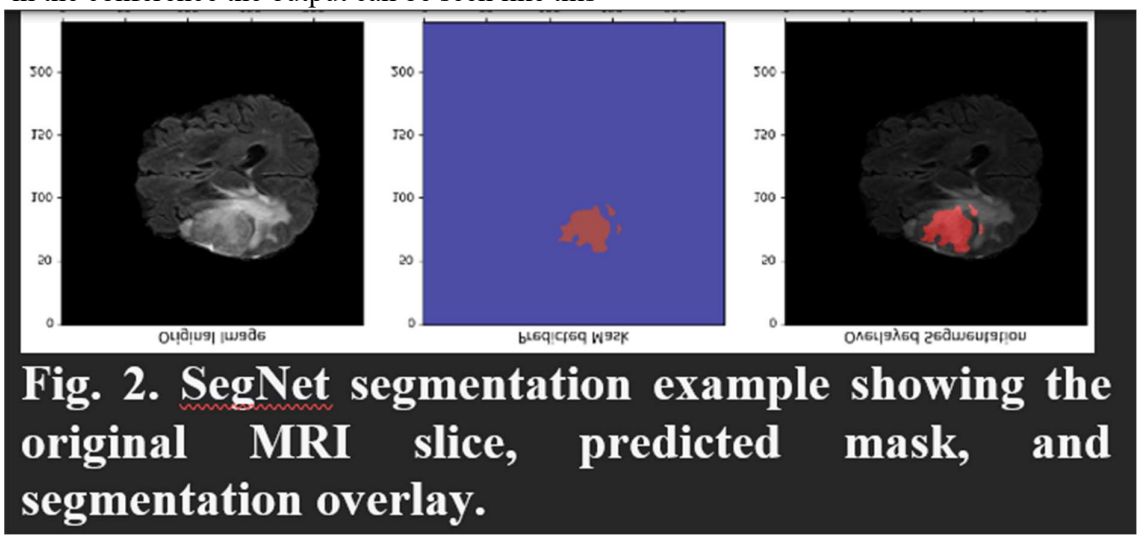
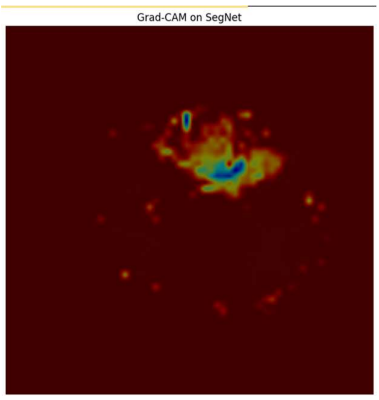
MODIFICATIONS REQUIRED

The first main query I have in the existing project is

- in the conference the output can be seen like this



Here in the actually project only u-net segmentation and grad cam can only be seen as output kindly add these features properly to the project
In the image the text is also inverted and it is not proper so kindly ensure that it is rectified and all the content in the conference matches with the project which is actually implemented.



here it is showing like grad cam on segnet make sure that the project is properly using that feature because in the project it shows simple gradcam heatmap.

Instead of streamlit try using some other frontend like tailwind with good userfiendly and interactive UI features.

NO 1:

Classification which was earlier mentioned is not added

- Classification of the tumour which type of tumour it is and what stage it is ….like malignant or not
- Whether it is serious or in the starting stage

NO 2:

Explainable AI or chatbot

- It explains the person about the tumour identified and how it has predicted the tumour reasons and explanation

NO 3:

Radiomics Dashboard

PyRadiomics already extracts:

- shape features
- intensity features
- texture features
- image diagnostics

from the segmented region.

But the UI doesn't show them.

Add a "Quantitative Analysis" tab

For example:

QUANTITATIVE TUMOR ANALYSIS

Tumor Area	12.43 cm²
Estimated Region	8.21 %
Mean Intensity	147.32
Intensity Variance	32.14
Texture Entropy	4.82
Compactness	0.71
Sphericity	0.64

Then display charts:

Texture Characteristics

Entropy 4.82

Contrast 3.21

Energy 0.42

Homogeneity 0.76

NO 4:

Actually use your SegNet + U-Net ensemble

This is a **big missed opportunity**.

Your research code already calculates:

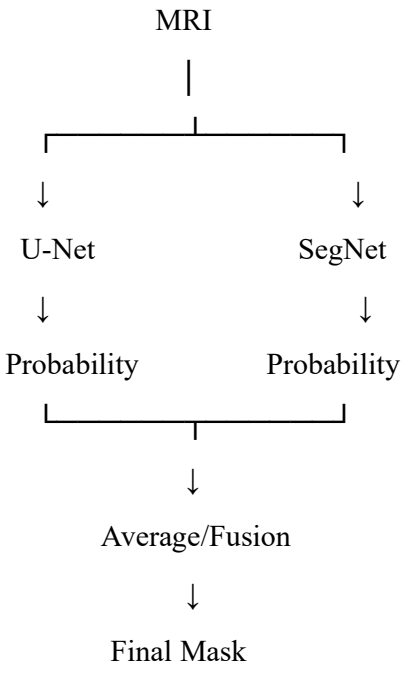
$$P_{ensemble} = \frac{P_{SegNet} + P_{UNet}}{2}$$

according to the project document.

But the application isn't using SegNet during inference.

So instead of:

U-Net → Mask make it:



Then the UI can show:

Model Comparison

Model	Result
U-Net	Segmentation
SegNet	Segmentation
Ensemble	Final segmentation

And add:

U-Net Prediction

SegNet Prediction

Ensemble Prediction

side-by-side.

This gives you a legitimate architectural contribution:

Probability-level ensemble of complementary segmentation networks

rather than simply saying "we used U-Net."

NO 5:

Add uncertainty estimation

This is one feature I would **strongly consider** because medical AI should not simply say:

"Tumor detected."

Instead:

Prediction

———————————

Abnormal Region Detected

Confidence



Segmentation Confidence



NO 6:

Tumor/abnormality measurements

From the segmentation mask, calculate quantitative measurements.

For a 2D image you can safely report things like:

Region measurements

- pixel area
- bounding box
- width
- height
- percentage of image occupied by predicted region
- centroid
- perimeter
- shape descriptors

For example:

ABNORMAL REGION ANALYSIS

Detected Region: Yes

Region Area: 18,426 pixels

Image Coverage: 7.02%

Bounding Box:

Width: 143 px

Height: 128 px

Centroid:

X: 341

Y: 267

NO 7:

Automatic medical-style report generation

This would make the demo **look dramatically more complete**.

After analysis:

MRI ANALYSIS REPORT

Patient/Scan ID: MRI_001

Segmentation

Abnormal region detected

Predicted Region Coverage: 7.02%

Quantitative Features

Mean Intensity: 147.32

Entropy: 4.82

Contrast: 3.21

Sphericity: 0.64

Model Prediction

Class: Abnormal

Confidence: 87%

Explainability

Grad-CAM generated

Then:

Download Report

You could generate a PDF.

The report should clearly say something like:

Research prototype — not intended for clinical diagnosis.

That's important for a medical-AI student project.

NO:

If possible try to add a database